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REGULAR PAPER

# Illu-NASNet: unsupervised illumination estimation based on dense spatio-temporal smoothness

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**Abstract**
Illumination estimation is a highly challenging problem. Many methods are learning from multi-image by unsupervised learning. However, these methods are based on unrelated data and do not exploit the information between images. In this paper, we propose a new unsupervised method for learning illumination by observing indoor video sequences under changing illumination and learning the intrinsic decomposition of the sequences. This method allows us to learn without ground truth (GT) and leverage information obtained from multiple consecutive images. Based on the above ideas, we propose a new network framework and introduce albedo smoothness loss and illumination smoothness loss two new dense spatio-temporal smoothness loss functions. These loss functions take full advantage of the information between images to constrain the entire image sequence. In our evaluation, our approach shows good performance on several specific metrics. Experiments show that our method has strong generalization and can be easily applied to other classical datasets, including Intrinsic Images in the Wild (IIW) and Shading Annotations in the Wild (SAW).

**Keywords** Illumination estimation · Unsupervised learning · Dense spatio-temporal smoothness · Dense CRF · Isotropic spherical Gaussian

## 1 Introduction

Image intrinsic decomposition(IID) is the question of decomposing input data $I$ into a product of an albedo map and a shading map: $I = A \times S$. As part of the image's intrinsic properties, location, direction, intensity, and illumination color significantly affect the image's appearance. Separating and replacing the original lighting can bring a three-dimensional effect to the image. Virtual objects can be rendered by adding lighting in Augmented Reality (AR) to be more realistic. While previous work made significant progress in a single-image intrinsic decomposition [9, 11], it is still a challenging problem. With the wide application of deep learning in the image field, many researchers try to use deep learning to solve this task [7, 20]. Unfortunately, supervised learning requires ground truth as labels, and collecting labels is extremely difficult. To solve this problem, many prior works using synthetic data or manual annotation [5, 7, 9, 17], but it is challenging to generalize synthetic data to natural scenes, and manual annotation methods do not entirely label the correct ground truth. Therefore, these methods are severely limited.

Instead, we consider a more readily available, alternative training data to learn the scene's illumination and other intrinsic information: We use a continuous image sequence from a video whose viewpoint and illumination are constantly changing and persistent. Based on the above considerations, we collect and make a constant image sequence dataset from videos. Each image sequence is continuous, and illumination is varied (e.g., the light switches on/off, and the color of light changes). Although the image sequences do not provide ground truth (GT), we can utilize their mutual

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relationship during training. Our approach bridges the gap between optimization-based strategies and deep learning approaches. Meanwhile, our new loss function includes the prior optimization-based methods and applies over the entire image sequence, as illustrated in Fig. 1.

Summarily, our contributions are as follows:

1. We propose a new network framework to predict illumination, surface normal, and albedo of shading (IllumNASNet).
2. We propose to generate a shading map from the surface normal map and illumination map. The illumination map is generated with fewer Gaussian parameters (72 parameters), making capturing information such as shadows and highlights easier.
3. We introduce two different dense spatio-temporal smoothness losses and show how to apply these losses in our network framework. Our method achieves competitive and superior results in our experiments on two classic real-world datasets, IIW [17] and SAW [5].

## 2 Related work

### 2.1 Illumination estimation based on optimization

Because illumination estimation is a complex and challenging problem, human-devised priors based on observation are used to regularize optimizations. For example, the first proposed Retinex algorithm [6] assumes that the albedo image has a large gradient in the image rather than an illumination image. Meanwhile, many different priors and optimization tools are proposed to guide the illumination prediction, such as inference illumination from conditional random fields (CRFs) [17]. Some recent researchers use single image depth information and surface normals [9, 11]. Although surface normal and depth information can improve illumination prediction, such methods

assume normal and depth information are available during optimization. Meanwhile, we believe that the method of learning from multi-image can make up for the disadvantage that illumination is difficult to capture in complex real-world scenes. Therefore, many researchers pay more attention to multi-image.

### 2.2 Illumination estimation from multi-image

Many methods estimate illumination and intrinsic information from multi-image by assuming constant albedo but varying illumination over time [4, 8]. The data they use is discontinuous in time, and the viewing angle is fixed, which leads to drastic lighting/shading changes from image to image, and does not utilize the illumination information from image to image. At the same time, the priors for these methods are derived from multi-image or complex light source detection. We find that in real applications, these methods rely on a) large amounts of training data and b) images captured outdoors in bright sunlight or a laboratory with continuously changing illumination. In contrast, our method is more realistic because our data collect from video sequences with varying illumination. The images are temporally continuous and the lighting/shadow changes between consecutive frames are small, which also verifies our idea: the lighting changes continuously between consecutive frames of images.

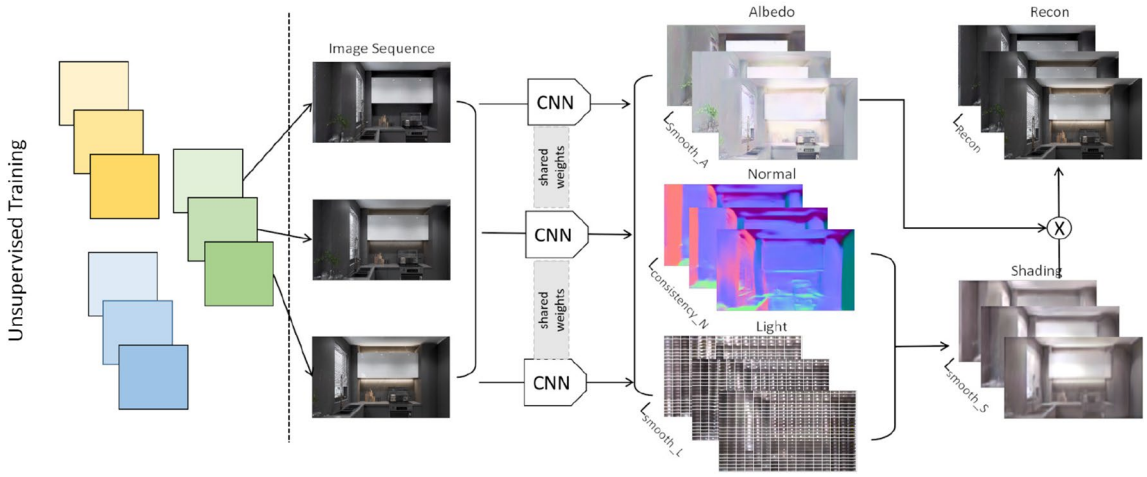


**Fig. 1** The network takes as input a time-varying sequence and predicts output consistent with the input sequence. Different loss functions are used to constrain the decomposition of predictions

### 2.3 Illumination estimation via unsupervised learning

Recently, several unsupervised approaches that use multi-image are introduced [8, 10], the consistency of scene variables in images being used to constrain the solution, especially the similarity in the albedo of the same object. Ma et al. [12] proposed an unsupervised learning-based method that takes a pair of images as input with different illumination and generates corresponding intrinsic prediction from a single image. However, this method only uses the constraint of albedo similarity, and input data is unrelated. So, its prediction results are not ideal. Instead, we input a set of temporally continuous image sequences and learn the decomposition under varying illumination conditions. This method exploits the temporal correlation of two consecutive images, which is not addressed in previous work.

## 3 Proposed method

In this section, we describe our new method and framework for learning albedo maps $A$, normal maps $N$ and illumination maps $L$ from our dataset, and introduce two new dense spatiotemporal smoothing loss functions Albedo smoothing loss and illumination smoothing loss, which make full use of the information between images to provide constraints for network training, we will describe these two losses in detail in Sect. 3.3.3 and Sect. 3.3.4.

### 3.1 Integrated lighting

Most of the previous work is to predict gray-scale shading. In this section, we recover the color shading image. We assume the same chromaticity $c$ of each pixel in the shading image $S$. In the Lambertian scene, the image can be decomposed into:

$$I_i = A_i \times S_i = c \times A_i \times S_i \tag{1}$$

where $A_i$ and $S_i$ represent the albedo map and shading map, respectively. Surface normal can help predict shaded images, so to use normal information, we generate shaded images via surface normals and global illumination maps. In complex indoor scenes, object surfaces are illuminated by incident light from multiple directions. Therefore, the shading of the illumination point $p$ is the sum of the incident illumination passing through the point $p$:

$$S = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} f_d(N, L) \sin\alpha d\alpha d\beta \tag{2}$$

where $N$ denotes the normal to a surface. The function $L(\alpha, \beta)$ expresses incident lighting passing through the point $(\alpha, \beta)$ on the upper hemisphere of the point $p$.

Inverse rendering for indoor scenes requires predicting spatially varying lighting for every pixel in the image. We use the pre-calculated radiation transmission method and the isotropic spherical Gaussians (SG) with fewer parameters. We use 12 spherical Gaussian lobes (72 parameters). The final illumination function $L(v)$ is defined as:

$$L(v) = w_k \sum_{k=1}^{n} G\left(v, d_k, \lambda_k\right), \quad G = e^{\lambda(1 - v \cdot d)} \tag{3}$$

where $d$ is the axis and direction, and it controls the position of the lobe on the sphere $S^2$, $w_k$ controls the amplitude or strength of the lobe, and $\lambda$ controls the clarity. Each spherical Gaussian lobe consists of 6 parameters $\left\{w_k, d_k, \lambda_k\right\}$. We sample $16 \times 8$ lighting directions, which is enough to restore most high-frequency lighting.

### 3.2 Architecture of illu-NASNet

Our proposed Illu-NASNet is shown in Fig. 1. It consists of two parts: albedo, normal estimation module and spatially varying lighting prediction that accounts for the global illumination effects such as shadow, highlights, etc.

We take a set of $m$ input images with $n$ pixels as a training instance. We input an image sequence $I = \left\{I^i \mid i = 1 \ldots m\right\}$ and predict the corresponding albedo, normal and lighting sequences by $A = \left\{A^i \mid i = 1 \ldots m\right\}$, $N = \left\{N^i \mid i = 1 \ldots m\right\}$ and $L = \left\{L^i \mid i = 1 \ldots m\right\}$. Additionally, for each image $I^i$, we have a binary mask $M^i$ indicating which pixels are valid.

The architecture of our network is shown in Fig. 2. Our network design is based on the U-Net frame [16]. The U-Net network is not only simple and efficient, but it is deep/shallow features have their meanings: the shallow network pays more attention to local features such as textures; the deep network pays more attention to those global features, and the U-Net network structure is achieved by feature splicing Realize the retrieval of edge features. Our network consists of one encoder and three decoders to predict albedo, normal, and lighting information separately, with feature concatenation between encoders and decoders. Each encoder layer consists of a $4 \times 4$ stride-2 convolutional layer, followed by group norm and Relu activation layers. Each layer performs on-channel feature fusion when upsampling. We add a fine-tuned network consisting of two convolution layers and two upsampling layers to optimize our regression task further.

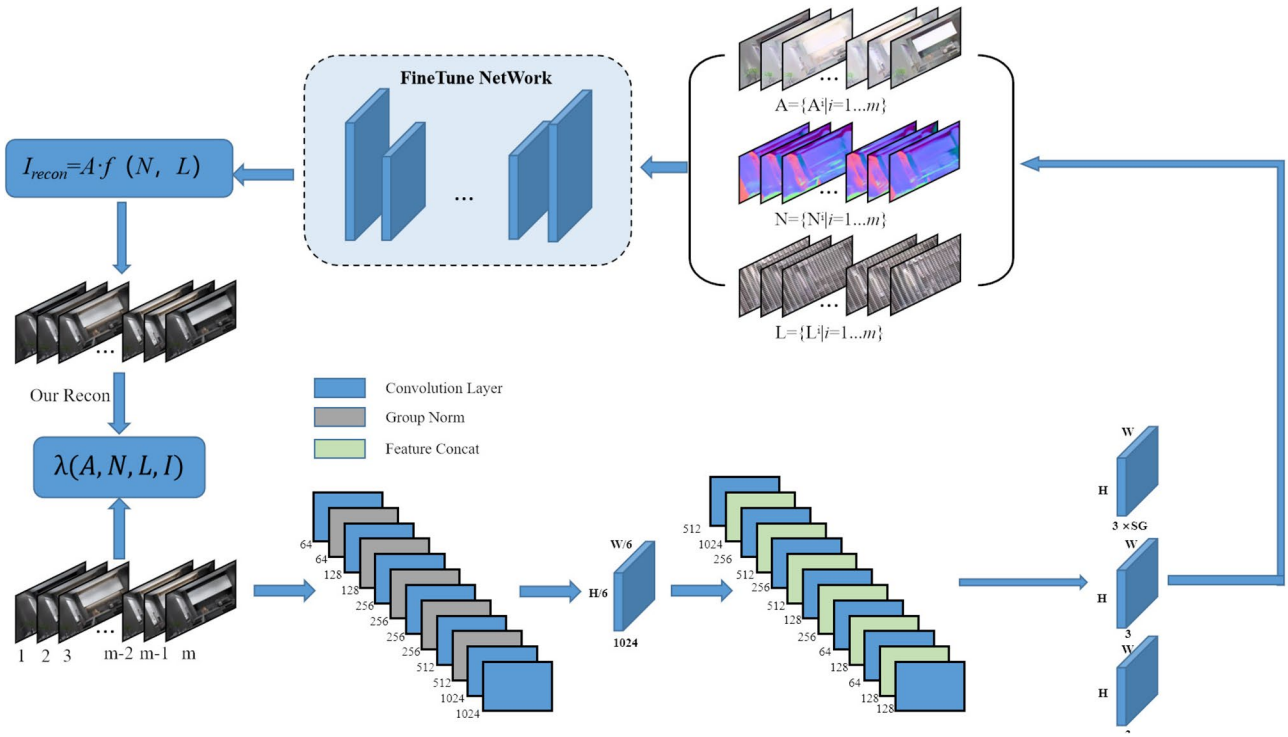


**Fig. 2** Network architecture. The Illu-NASNet uses an encoder-decoder structure with feature fusion at upsampling

### 3.3 Loss function

In unsupervised training, the loss function plays an extremely important role. We use the relationship between image sequences and self-constraint, and introduce two new dense spatio-temporal smoothness losses, albedo smoothness loss and illumination smoothness loss, This loss exploits the information between consecutive images for constraint. We use dense CRF to formulate our problem on the image sequence $I^m$, our target is maximize probability $p(A, N, L, I) = \frac{1}{\varphi}\exp(-\lambda(A, N, L, I))$ where $\varphi$ is the partition function. Maximizing $p(A, N, L, I)$ is equivalent to minimizing function $\lambda(A, N, L, I)$. So, we define this total loss function as:

$$\lambda(A, N, L, I) = w_0 L_{\text{Rec}} + w_1 L_A + w_2 L_{\text{Asmooth}} + w_3 L_{\text{Ismooth}} + w_4 L_S \tag{4}$$

Next, we describe each loss in Equation 4.

#### 3.3.1 Image reconstruction loss

During training, we input an image sequence $I^m$ and reconstruct single image $I^i \in I$ by the image forming model using the predicted albedo, normal and illumination map. Furthermore, we assume the same scene albedo is consistent, so we can use the albedo $A^j$, normal $N^i$ and illumination $L^i$ in the sequence to reconstruct the image $I^i$. Therefore, we can recover the reconstruction loss with all the images in the input sequence. The reconstruction loss is defined as follows:

$$L_{\text{Rec}} = \sum_{i=1}^{m}\sum_{j=1}^{m} \|I^i \otimes M^i \otimes M^j \otimes (\log I^i - \log A^j - \log N^i - \log L^i)\|_F^2 \tag{5}$$

where $\otimes$ is the Hadamard product. $M^i$ and $M^j$ are the binary mask of valid image region. We compute our loss in the log domain to simplify the computation, and image differences in dark regions are magnified in log space.

#### 3.3.2 Albedo consistency loss

Based on the albedo consistency assumption under the same scene, we introduce an albedo consistency loss that assumes the reflectance is approximately identical in the whole input sequence. The predicted map of albedo $A$ is unsupervised by:

$$L_{\text{A}} = \sum_{i=1}^{m}\sum_{j=1}^{m} \left\|M^i \otimes M^j \otimes \left(\log A^i - \log A^j\right)\right\|_F^2 \tag{6}$$

#### 3.3.3 Albedo smoothness loss

Based on the assumption that the albedo is approximately equal in the same scene, we define the albedo smoothing term $L_{\text{smooth A}}$. We train a continuous sequence of images, so we can define an albedo smoothing term involving all images that constrain all albedo images in the sequence. We propose a new dense spatio-temporal albedo smoothing loss based on the above ideas. The albedo smoothing loss is defined as:

$$L_{\text{Asmo}} = \sum_{\substack{i=1 \\ i \in I^m}}^{m} \sum_{\substack{p,\hat{p}=1 \\ p,\hat{p} \in I}}^{n} W_{p\hat{p}} \left( \log R_p^i - \log R_{\hat{p}}^{i+1} \right)^2 \tag{7}$$

where $p$ and $\hat{p}$ indicate pixels in the image sequence, and $W$ is a covariance matrix capturing the affinity between any two pixels $p$ and $\hat{p}$.

#### 3.3.4 Illumination smoothness loss

Based on the inspiration of the optical flow method, we propose a dense spatio-temporal illumination smoothing loss term, the illumination intensity at the same position in consecutive video frames changes gradually, as shown in Fig. 3. To consider the spatio-temporal information between images, it is a requirement that the input data must be continuous in time. The loss term is defined as:

$$L_{\text{Ismo}} = \frac{1}{2} \sum_{p=1}^{n} \sum_{i=2}^{m} \left\| L_p^{i+1} + L_p^{i-1} - 2L_p^i \right\|_F^2 \tag{8}$$

where $p$ indicates pixels in the same image position, and $i$ represents the image sequence number and its initial value is $i = 2$.

#### 3.3.5 Shading loss

Since the shading map is the dot product of surface normal and illumination, the shading loss is defined as:

$$\begin{cases} L_{\text{s}} = \sum_{i=1}^{m} \left\| \left( \log I - \log \widehat{A}_i \right) - \log \widehat{S}_i \right\|_F^2 \\ + w_s L_{\text{Ismo}} \\ \widehat{S}_i = \left\langle \widehat{N}_i, \hat{L}_i \right\rangle \end{cases} \tag{9}$$

where $\mathcal{W}_s$ is a weight matrix, $<,>$ is the dot product of normal and illumination.

## 4 Experiments

### 4.1 Implementation details and metrics

Our Illu-NASNet is implemented using PyTorch. We collected whole 120 image sequences with 7000 images for training. We augment the data by randomly rotating, stretching, and cropping. We reset the image to 256 $X$ 256 size for the image input and used the Adam optimization algorithm to train our network. For the training, we use the batch size of 16 and the training epochs of 14. We implement our method on an A100-SXM4-40GB GPU graphics card. The average decomposition time per image is 50ms, and Illu-NASNet achieves 14 FPS. Please refer to the additional file for details on the network design and examples of our dataset.

We use the human perceptual error metric proposed on the IIW dataset for reflectivity evaluation, representing the Weighted Human Disagreement Rate (WHDR). Since the SAW dataset provides some labels by manual annotation, we can use this information for evaluation. For the evaluation of the shading images, we use the method proposed

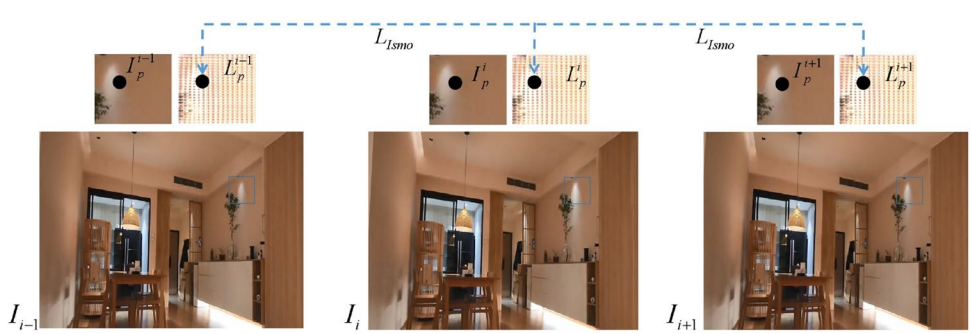


**Fig. 3** Illumination smoothness loss

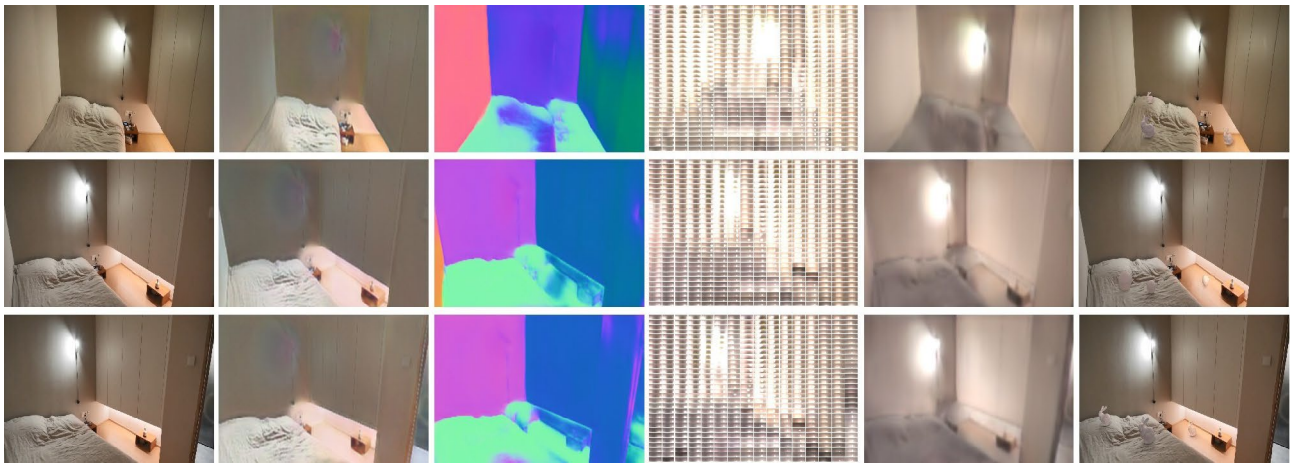


**Fig. 4** Testing results on our dataset. We estimate albedo, normals, light and shading. These are used for object insertion

**Table 1** Ablation study

| Method | Training set | WHDR(%) | AP(%) |
|---|---|---|---|
| w/per-imageLrec | Ours | 21.2 | 89.07 |
| w/local Lrsmo | Ours | 23.4 | 96.98 |
| w/grayscale $S$ | Ours | 20.3 | 95.43 |
| full method | Ours | 19.7 | 97.74 |

by Li and Snavely [8], who evaluate the shading map in the logarithmic domain. In addition, Li and Snavely also proposed challenge precision (P(c)) and unweighted precision (P(u)) [8]. We also use these two methods to evaluate our results. Further, we compute the average unweighted accuracy (AP(u)) and average challenge accuracy (AP(c)) for surface normal and shading.

To take both precision and recall into account, Luo et al [11] introduce the maximum $F_1$ score:

$$MF_1 = \mathrm{Max}_t\left(2 \times \frac{\mathrm{Precision}_t \times \mathrm{Recall}_t}{\mathrm{Precision}_t + \mathrm{Recall}_t}\right) \tag{10}$$

## 4.2 Ablation study

We make different configurations of the network. First, we replace the reconstruct loss that considers all images with a single-image reconstruction loss. Second, for albedo estimation, we replace our proposed dense spatio-temporal smoothness loss with pixel similarity between adjacent pixels. Finally, we replace the color shading image with grayscale shading map to evaluate the effect of color on the shading image. The specific experimental results are shown in Table 1.

**Table 2** Numerical evaluation results on our dataset

| | Albedo | | Normal | |
|---|---|---|---|---|
| Method | MSE | LMSE | Mean | Median |
| Nestmeyer [14] | 0.0149 | 0.0169 | 43.6 | 43.0 |
| BigTime [8] | 0.0116 | 0.0135 | – | – |
| Inverserendernet [19] | 0.0158 | 0.0124 | 40.0 | 38.0 |
| Yu et al [18] | 0.0112 | 0.0128 | 31.9 | 35.7 |
| Our approach | **0.0082** | **0.0093** | **30.2** | **28.9** |

## 4.3 Evaluation

### 4.3.1 Evaluation on our dataset

We evaluate our performance on our dataset. The network is trained on our dataset, and we show our test results on this dataset in Fig. 4. We evaluate the numerical evaluation results of the albedo and normal of the multi-image method on our dataset separately. Experimental results show that our method outperforms all current multi-image illumination estimation methods. The numerical comparison results are shown in Table 2, the last line in bold indicates the optimal result of the method in this paper.

### 4.3.2 Evaluation on IIW and SAW

P(u) and P(c) are used to evaluate surface normal and shading, WHDR is used to evaluate albedo. Experimental results show that our unsupervised model achieves competitive results in shading and albedo estimation compared to models trained on synthetic datasets [7, 21]. Compared with

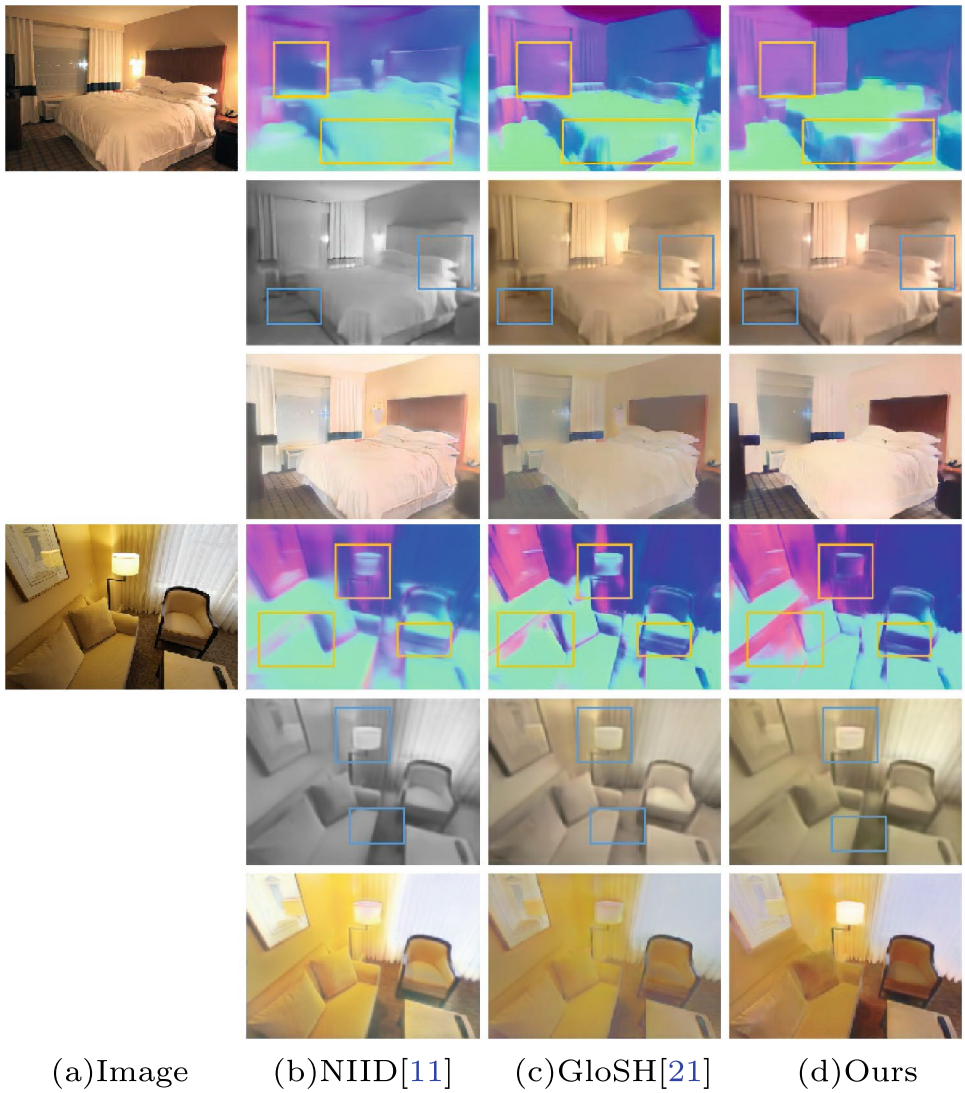


**Fig. 5** Qualitative comparisons results on the NYUv2

the network trained on real datasets, the Illu-NASNet also obtained excellent results, as shown in Table 3.

Figure 5. shows visible results that use surface normal. Our method predicts shading maps have more correct textures and more pronounced variations in lighting and shadows than GLosh [21]. Compared with the method proposed by NIID [11], our shading smoothness achieves the same effect as theirs, but their disadvantage is that the network design is complex, and the training takes a lot of time. Our network design is lighter-weight and adds additional constraints to the entire image sequence. It can be seen from the ablation study that if the local smoothing loss is used, the final model does not reach our expected results.

More visual comparisons are shown in Fig. 6. The qualitative results from our method and two other state-of-art algorithms, Li et al. [9] and Luo et al. [11], on test images from IIW and SAW. Our results are visually comparable to these methods.

#### 4.3.3 Object insertion comparison

In addition to evaluating the experimental results numerically, the visual effects of illumination predictions are also important. The virtual object insertion application visually demonstrates the consistency of the predicted and original scene illumination. We compare our method with previous methods [1, 4, 9], as shown in Fig. 7. The insertion results show that we restore consistent illumination and render

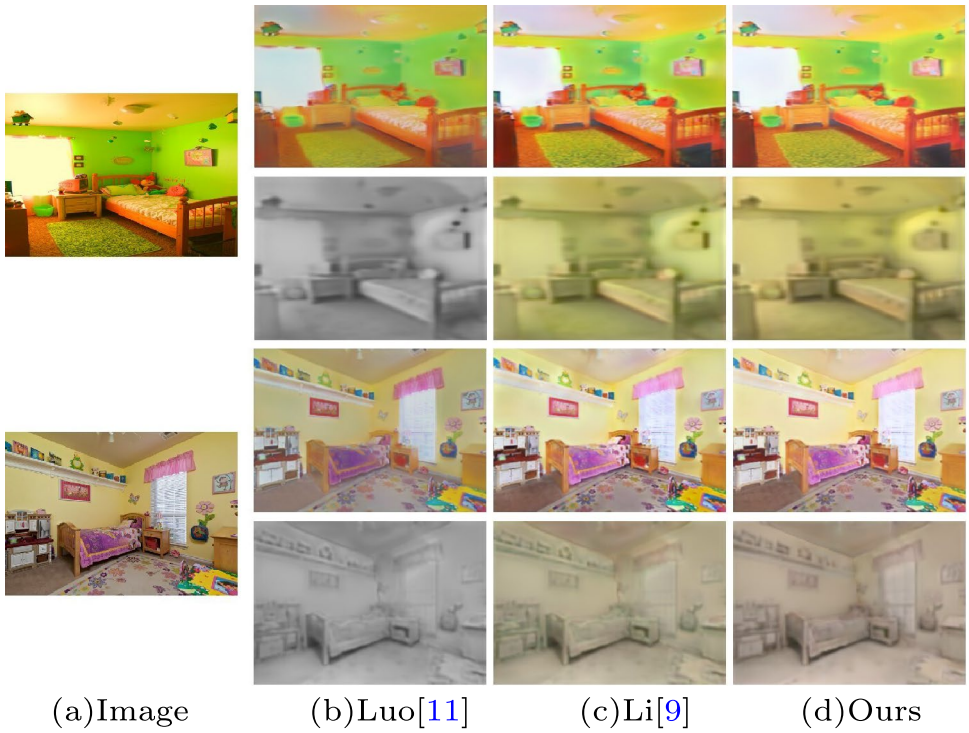


**Fig. 6** Qualitative comparisons results on IIW/SAW

**Table 3** Numerical evaluation results on the SAW and IIW

| Method | Training set | WHDR(%)(↓) | P(u) | | P(c) | |
|---|---|---|---|---|---|---|
| | | | AP(%) | mF1 | AP(%) | mF1 |
| Zhao et al. [15] | – | 23.8 | 97.11 | 0.909 | 89.72 | 0.795 |
| Garces et al. [3] | – | 24.8 | 96.89 | 0.906 | 92.39 | 0.823 |
| Bell et al. [17] | – | 20.6 | 97.37 | 0.913 | 92.18 | 0.834 |
| Fan et al.* [2] | IIW | 14.5 | – | – | – | – |
| Zhou et al.* [22] | IIW | 19.9 | 96.24 | 0.898 | 86.34 | 0.759 |
| Gehler et al. [13] | IIW | 19.5 | 97.26 | – | 89.94 | – |
| Li et al.* [7] | CGI | 17.8 | 98.39 | 0.935 | 94.05 | 0.849 |
| GLoSH [21] | SUNCG | 26.8 | – | – | 92.40 | – |
| Snavely et al.* [8] | BigTime | 20.3 | 97.03 | – | 88.64 | – |
| Li et al. [9] | OpenRoom | 21.9 | 95.09 | 0.923 | – | – |
| Liu et al. [10] | CGI+IIW | 18.9 | – | – | 90.57 | 0.883 |
| Yu et al. [18] | MD+BT+NYU | 21.1 | 92.56 | 0.837 | – | – |
| Our | Ours | **18.7** | **97.74** | **0.941** | **93.18** | **0.890** |

The last line in bold indicates the optimal resultof the method in this paper

*Predictions of the network have been post-processed

shadows on virtual objects, achieving the same effect as the illumination of objects in the real scene.

## 5 Conclusion

We propose Illu-NASNet, an unsupervised framework that observes image sequences with varying illumination over time. Our method obtained competitive results from prior works that focus on sub-problems, which highlights the impact of our dataset, loss function and network design. We present object interpolation results on real images. The global illumination is captured by our per-pixel SG lobe. We approximate the constraints of albedo consistency. In future work, we will try to estimate illumination for the same scene from different viewpoints using normal and albedo alignment.

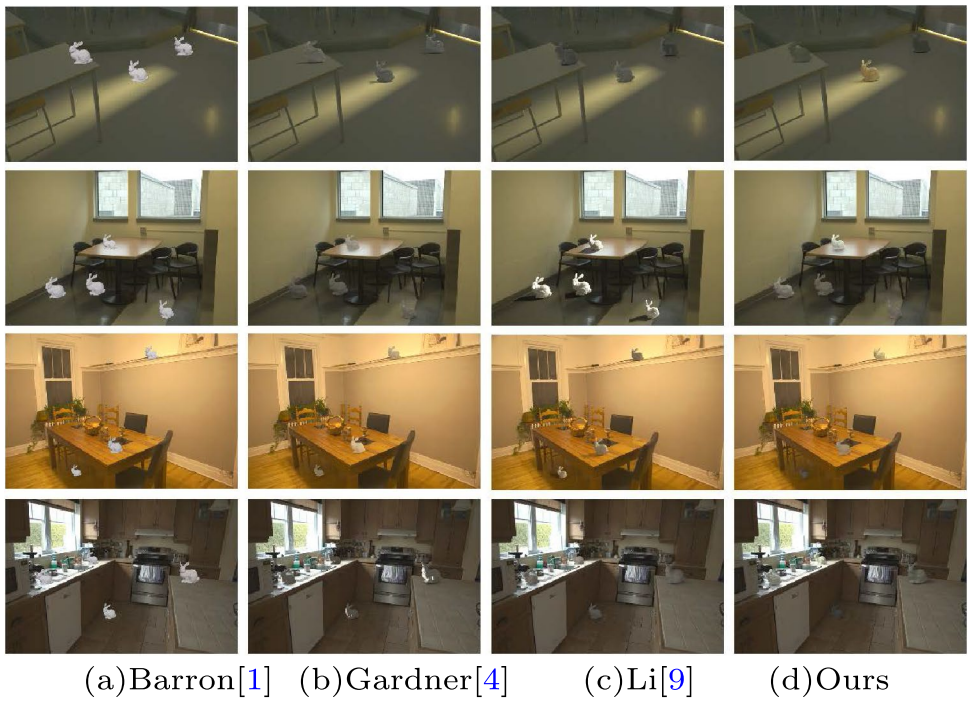


**Fig. 7** Object insertion examples and comparisons

**Supplementary Information** The online version contains supplementary material available at https://doi.org/10.1007/s00530-023-01057-2.

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**Author Contributions** This idea was proposed by ZZ, who experimented, data analyses and wrote the manuscript; CH, RH, YL and YC gave necessary help to this work and put forward valuable opinions.

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**Availability of data and materials** The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

## Declarations

**Conflict of interest** The authors have no relevant financial or non-financial interests to disclose.

**Ethical approval and consent to participate** Not applicable.

**Consent for publication** Informed consent was obtained from all individual participants included in the study.

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